NEURODETECT

Technical Design & System Architecture

MRI-Based Alzheimer’s Disease Classification Using a Hybrid CNN–FFNN Architecture

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PROJECT SNAPSHOT

Input	2D MRI image files associated with the OASIS dataset
Feature extractor	ResNet50 convolutional neural network (CNN)
Classifier	Feedforward neural network (FFNN) with Softmax output
Output	Four image classes representing dementia status/severity categories

This document describes the project pipeline shown in the supplied architecture diagram and the implementation details available for the project. Items that depend on the exact code or a reproducible final run are identified rather than assumed.

1. Project Overview

NeuroDetect is a research-oriented system for classifying MRI images into four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The design uses a hybrid pipeline: a ResNet50 CNN extracts image features, and a feedforward neural network processes those features to produce a class prediction.

The intended output is an automated classification result for research and educational evaluation. It should not be described as a clinically validated diagnostic tool unless independent clinical validation has been performed.

2. System Architecture

The supplied architecture diagram represents the workflow from MRI image input through preprocessing, feature extraction, classification, and the final four-class output.

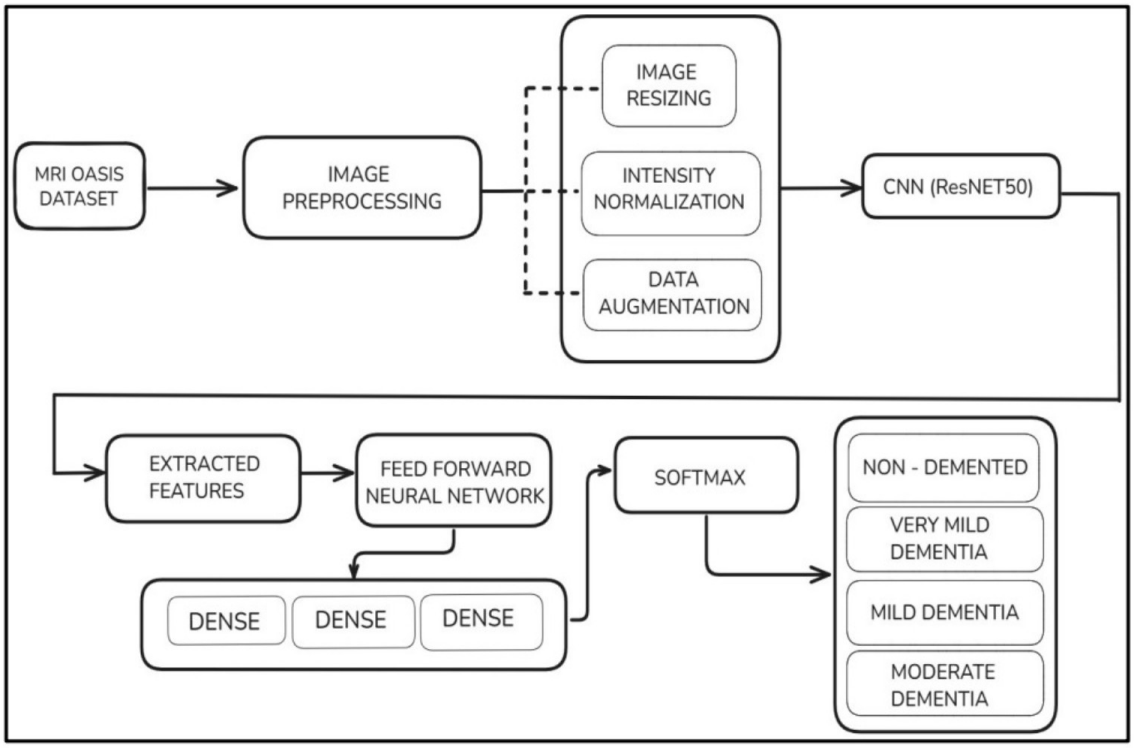


Figure 1. Supplied NeuroDetect architecture: MRI OASIS dataset → preprocessing → ResNet50 CNN → extracted features → FFNN → Softmax → four class labels.

3. Input Data and Preprocessing

The project uses 2D JPEG MRI image slices associated with the OASIS dataset, rather than describing the current pipeline as a direct 3D NIfTI-volume workflow. The preprocessing stage prepares images for the feature extractor.

Stage	Purpose
Image resizing	Resizes images to the model input size; the documented implementation uses 224 × 224 pixels.
Intensity normalization	Scales image values into a consistent numerical range for model input.
Data augmentation	Training-time transformations may include rotation, translation, horizontal flipping, and brightness/contrast variation, as configured in the implementation.

4. ResNet50 Feature Extraction

ResNet50 is used as the CNN feature extractor. Its convolutional layers learn visual representations from the preprocessed MRI images. The project design describes the extracted representation as a 2,048-dimensional feature vector, which is passed to the downstream FFNN classifier.

Why this stage is used

Separating feature extraction from classification provides a clear modular design: the CNN converts image pixels into a feature representation, while the FFNN maps that representation to one of the target categories.

5. Feedforward Neural Network (FFNN)

The FFNN receives the feature vector generated by ResNet50. The described architecture uses dense layers with nonlinear activations and dropout regularization, followed by an output layer with four class scores. Softmax converts the output scores into a probability distribution over the four classes.

Component	Role
Dense hidden layers	Learn combinations of extracted image features for the classification task.
ReLU activation	Introduces non-linearity in the hidden layers.
Dropout	Regularization intended to reduce overfitting during training.
Four-unit output layer	Produces a score for each target class.
Softmax	Converts class scores into normalized class probabilities.

6. Output Classes

Class label	Meaning in the project
Non-Demented	MRI image assigned to the non-demented category.
Very Mild Demented	MRI image assigned to the very-mild category.
Mild Demented	MRI image assigned to the mild category.
Moderate Demented	MRI image assigned to the moderate category.

7. Training and Evaluation Design

The project design uses a training, validation, and test workflow. A subject-level split is intended to reduce the risk of images from the same individual appearing across different subsets. The exact split counts and final evaluation figures should match the reproducible run used for the published results.

Training configuration to document

• Input size: 224 × 224 pixels.

• Optimizer: Adam.

• Learning rate: 0.0001.

• Batch size: 32.

• Loss: categorical cross-entropy for the four-class task.

• Evaluation: accuracy, per-class precision/recall/F1, macro and weighted averages, and a confusion matrix.

Only report numerical results that come from the final model run and can be reproduced. If a confusion matrix or metrics were illustrative rather than generated by the trained model, they should not be presented as measured results.

8. Implementation and Reproducibility Notes

• Keep the dataset outside the public repository unless its license and sharing terms explicitly allow redistribution.

• Record package versions and provide a requirements.txt file for a repeatable environment.

• Document the actual training and evaluation commands, file paths, and expected outputs.

• Use a fixed random seed where supported and record the subject-level split used for evaluation.

• Keep model checkpoints and large generated artifacts out of Git unless they are intentionally published and appropriately licensed.

9. Limitations and Future Work

Potential limitations include class imbalance, limited subject diversity, and the difference between classifying individual 2D slices and assessing a complete clinical MRI study. Future work may investigate more balanced training, independent external evaluation, additional subject diversity, and 3D MRI approaches if suitable data and implementation resources are available.

10. Conclusion

NeuroDetect combines ResNet50-based CNN feature extraction with an FFNN classifier to explore four-class classification of MRI images associated with Alzheimer’s disease. Its modular design separates image preparation, feature learning, and final classification. Reproducible testing and independent validation are necessary before drawing conclusions about generalization or any clinical usefulness.

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